13. Monday, April 28 Mathematical Induction (4.1, 4.2)

BUT before, recall from last time: Exercise: Read uploaded proof of Thm 6 on Lecture 12.

- **Division Theorem**: For any $a \in \mathbb{Z}$ and $n \in \mathbb{N}$, there are unique $q, r \in \mathbb{Z}$ with
$$a = nq + r \quad \text{and} \quad 0 \leq r < n.$$
(q: Quotient, r: Remainder)

- **Corollary**: There is a unique $r \in \mathbb{Z}$ with $0 \leq r < n$ such that $a \equiv r \pmod{n}$

Theorem 1: For any integer n, we have $n^3 \equiv n \pmod 3$

Proof: Fix $n \in \mathbb{Z}$. By the Division Theorem, there are three possibilities for the remainder of n modulo 3:

- **Case 1**: $n \equiv 0 \pmod 3$. If $n \equiv 0 \pmod 3$ then $n^3 \equiv 0^3 \equiv 0 \equiv n \pmod 3$ (By Thm 12.2)
- **Case 2**: $n \equiv 1 \pmod 3$. If $n \equiv 1 \pmod 3$ then $n^3 \equiv 1^3 \equiv 1 \equiv n \pmod 3$
- **Case 3**: $n \equiv 2 \pmod 3$. If $n \equiv 2 \pmod 3$ then $n^3 \equiv 2^3 \equiv 8 \equiv 2 \equiv n \pmod 3$

In any case, $n^3 \equiv n \pmod 3$, as desired $\square$

Mathematical Induction

Theorem (Mathematical Induction)

Let $P(n)$ be an open sentence with universe $n \in \mathbb{N}$.

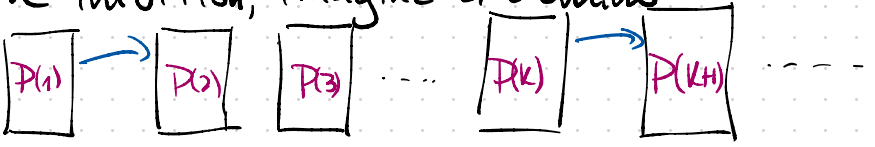
Suppose that: 1) $P(1)$ is true

2) For any $k \in \mathbb{N}$, if $P(k)$ is true then $P(k+1)$ is true.

Then $P(n)$ is true for any $n \in \mathbb{N}$

"Proof:" Can't provide a proof. mostly because we don't have a formal definition of $\mathbb{N}$

But, to gain some intuition, imagine a domino



Where the k-th piece falls if $P(k)$ is true.

Then (1) says that the first piece falls

(2) says that whenever kth falls then also (k+1)th falls

So it makes sense to deduce that all pieces will drop. "□"

Technique (Proof by Induction)

We prove that a statement $P(n)$ is true for any $n \in \mathbb{N}$ as follows:

1) **Base case**: Prove that $P(1)$ is true

2) **Inductive step**: Prove that, for any $k \in \mathbb{N}$, the conditional $P(k) \longrightarrow P(k+1)$ is true.

↳ Here we need to assume $P(k)$ true and need to show that $P(k+1)$ is true

Theorem 2: For any natural number n, we have $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

E.g. $P(5)$ states that $1+2+3+4+5 = \frac{5 \cdot 6}{2}$

Proof: For each $n \in \mathbb{N}$, let $P(n)$ denote the statement $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

We will prove by induction on n that $P(n)$ is true for all $n \in \mathbb{N}$.

Base case: We want to prove $P(1)$ is true, i.e. $\sum_{i=1}^{1} i = \frac{1 \cdot 2}{2}$. Both sides are equal to 1, so the base case holds.

Inductive step. We want to prove that, for any $k \in \mathbb{N}$, $P(k) \longrightarrow P(k+1)$.

Fix $k \in \mathbb{N}$ such that $P(k)$ is true, so know that $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$. "Inductive Hypothesis"

We wish to prove that $\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$

So we compute

$$\sum_{i=1}^{k+1} i = \left(\sum_{i=1}^{k} i\right) + k+1 \underset{IH}{=} \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+2)(k+1)}{2}$$

Which says that $P(k+1)$ is true.

Hence, by induction, $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ for all $n \in \mathbb{N}$ □

Back to Theorem 1:

Theorem 1: For any $n \in \mathbb{N}$, $3 \mid (n^3 - n)$

Proof: Let $P(n)$ be the statement $3 \mid (n^3 - n)$. We prove by induction on n that $P(n)$ is true for all $n \in \mathbb{N}$.

Base case: We want to show that $3 \mid (1^3-1)$

But $1^3-1 = 1-1 = 0 = 3 \cdot 0$, so by definition of divisibility, $3 \mid (1^3-1)$.

Inductive step: We want to prove that, for any $k \in \mathbb{N}$, the conditional $P(k) \longrightarrow P(k+1)$ is true.

Fix $k \in \mathbb{N}$ such that $P(k)$ is true. We know that $\boxed{3 \mid (k^3-k)}$ Inductive Hypothesis

So there exists some $m \in \mathbb{Z}$ such that $k^3-k = 3m$.

We want to prove $P(k+1)$ is true, which means $3 \mid ((k+1)^3-(k+1))$

Now we compute

$$\begin{aligned}(k+1)^3-(k+1) &= k^3+3k^2+3k+1-(k+1)\\ &= k^3+3k^2+2k\\ &= (k^3-k)+3k^2+3k\\ &\overset{IH}{=} 3m+3k^2+3k\\ &= 3(m+k^2+k)\end{aligned}$$

And $m+k^2+k \in \mathbb{Z}$ by closure. Hence $3 \mid ((k+1)^3-(k+1))$, i.e. $P(k+1)$ is true

Thus, by induction, $3 \mid (n^3-n)$ for any $n \in \mathbb{N}$ $\square$